Lab Report

Course title: Software Engineering Laboratory
Course code: CSE-3206
Session: July - Dec 2026

Date of Submission: *27th August, 2026*

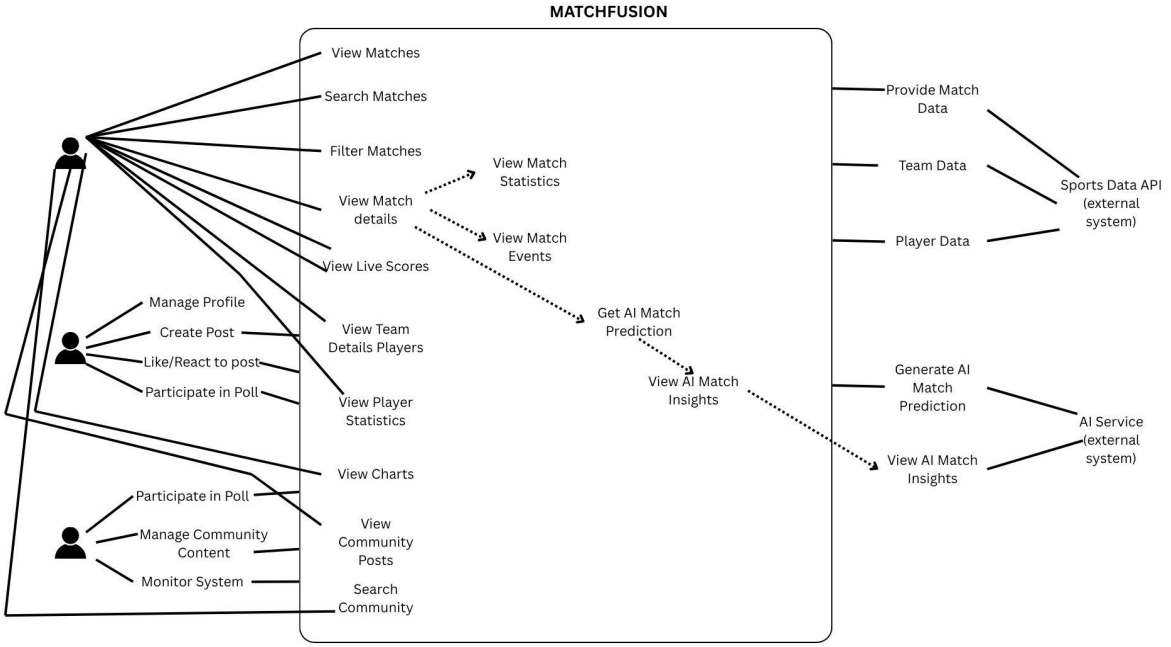
Submitted to-

Rumana Yasmin
Department of Computer Science and Engineering
Bangladesh University of Professionals

Tanjim Taharat Aurpa
Department of Data Science and Engineering
Gazipur Digital University

Team Members	ID	Name
01	24524203046	MD. Mehedi Hassan
02	24524203050	Tabassum Binte Tariq Rabsa
03	24524203112	Intasar Mostafiz
04	24524203170	Pritoma Paul Lopa

USER CASE DIAGRAM:



System Overview and Primary Actors

The MatchFusion Use Case Diagram illustrates the functional requirements and architectural interactions of the MatchFusion platform—an AI-powered sports community and match analysis system. The diagram is framed within a central system boundary enclosing all core functionalities, with user roles (primary actors) positioned on the left and external integration systems (secondary actors) positioned on the right. The system caters to three main categories of human users: unauthenticated Visitors/Fans, Registered Users, and System Administrators, each possessing escalating levels of authorization and interaction within the ecosystem.

User Roles and Interaction Scope

The Visitor or Fan represents the default primary actor seeking information without requiring full platform credentials. Users in this role can perform extensive data-viewing actions, including viewing live match lists, searching, and filtering matches. They can also inspect comprehensive match details, view live real-time scores, examine team details and player rosters, view individual player statistics, analyze visual charts, and browse or search through public community posts.

Building upon the base capabilities of a Visitor, the Registered User actor retains all viewing privileges while introducing interactive community and personalization features. Authenticated users can manage their personal profiles, draft and publish original community posts, engage with existing content by liking or reacting, and participate in interactive user polls.

At the administrative level, the Administrator actor focuses on platform management and governance. Admins maintain full operational oversight by monitoring system health,

moderating and managing user-generated community content to ensure compliance, and overseeing poll management.

Specialized Relationships and AI Workflow Integration

Within the core platform, several specialized use cases expand upon the primary "View Match Details" workflow. When a user views match details, this action logically extends to viewing detailed match statistics and specific match event timelines. Furthermore, viewing match details initiates a key automated process: retrieving AI match predictions. This prediction request seamlessly triggers a secondary process to generate and display AI match insights, providing users with deeper analytical context directly within their viewing session.

External System Dependencies

On the right side of the diagram, the system connects to two critical external systems that power its core data and intelligence features. The Sports Data API serves as the primary external data source, providing live match data, team structures, and detailed player performance metrics directly into the MatchFusion system. Complementing this, an external AI Service (such as an LLM or predictive engine) handles complex analytics by generating AI match predictions and processing predictive algorithms to deliver real-time AI match insights back to the platform interface.

SDLC Selection Matrix

Priority	Criteria	Waterfall	Agile
1	Clear, Fixed Requirements	Yes	No
5	Flexible and Adaptive	No	Yes
5	Continuous Feedback	No	Yes
3	Rapid Delivery	No	Yes
Total- 14	Over all	1	13